MAKE THINGS HAPPY !

IoTextra Octal

Digital I/O and Relays Module

The **IoTextra Octal** (Rev. 3.04) is an eight-channel digital I/O module, containing four digital input, two digital output channels and two relays.
The maximum digital input signal level is 36VDC.
Digital output signals are generated by NPN transistors. Transistor parameters include: collector current 8A (peak up to 16A), voltage up to 100VDC, and power dissipation 20W. The module can be used in extreme conditions only with appropriate cooling, including radiators. If SMAJ48A TVS diodes are installed to protect against transient processes, the maximum voltage is up to 48VDC.
Module has two [Panasonic APAN3105], or [Hongfa HF49FD] relays. Contact form – **SPST (1 Form A)**.
All digital channels have individual galvanic isolation with a dielectric strength of 3750 Vrms.
The module provides indication of the digital input and output channel status.
Up-to-date technical information about the module is available on GitHub.

SOFTWARE POWER:

The **IoTextra Octal** module offers native support for popular IoT platforms:

This hardware compatibility unlocks the full potential of the IoTflow platform. It enables you to:

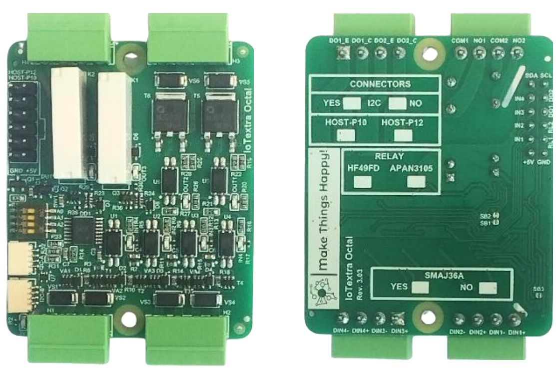
- Create custom **Node-RED** nodes effortlessly.
- Integrate via **MQTT** with third-party global platforms like **Blynk**.
- Scale your project from a single node to a complex industrial network.

Integrate with **Tasmota** for reliable **MQTT**-based telemetry and remote control — support for **IoTextra** modules, including **IoTextra Octal**, is described here.

There are two modes for using the **IoTextra Octal** module:

- ***GPIO Mode***. In this mode, reading the state of the input signals and controlling the output signals are performed using the **AP0-AP7** signals on the **HOST-P** connector.
- ***I^{2}C Mode***. In this mode, reading the state of input signals and controlling output signals are performed via the **I^{2}C** bus using the I/O expander installed on the module (TCA9534 or a compatible chip). Up to 16 modules can be connected to one **I^{2}C** bus.

Combined Use: It is also possible to use the **AP0-AP7** signals for some channels through the **HOST-P** connector and the I/O expander for the remaining channels.



This module is designed for systems requiring a mix of high-speed logic signaling and high-power load switching.

1) **Industrial Automation (Conveyor Belt Control)**: Transistor outputs (DO) provide start/reverse signals to a Variable Frequency Drive (VFD), while the relays switch a signal tower (stack light) or an electromagnetic brake.
   - **Software**: **Node-RED** and **IoTflow** for implementing complex timing sequences and interlocked motor start/stop logic.
2) **IIoT (Machine Status Monitoring)**: Digital inputs monitor safety sensors (e-stops, light curtains), transistor outputs drive a local status panel, and relays provide galvanic isolation for the remote emergency shutdown circuit.
   - **Software**: **Tasmota** for reliable MQTT-based telemetry and real-time state reporting to a central dashboard.
3) **Smart Farming (Greenhouse Automation)**: Inputs monitor water levels and light intensity; relays activate irrigation pumps or heaters, while transistor outputs drive low-power actuators for ventilation flaps.
   - **Software**: **Node-RED** and **IoTflow** for flexible rule-based control according to time-of-day and external environmental sensor data.
4) **Smart Home (Entrance Access Control)**: Inputs are connected to gate position sensors (reeds) and doorbell buttons; relays operate the electric strike lock and exterior lighting, while transistor outputs trigger an alarm buzzer and LED status indicators.
   - **Software**: **Home Assistant** for seamless mobile app integration and "Coming Home" scene automation.

FEATURES:

- Compatibility with major microcontrollers
- Module power supply is 5VDC, which feeds a 3V3 regulator for powering the logic section
- Includes a 3V3 Indicator LED
- Protection against reverse power supply polarity
- 4 independent, optically isolated digital input signals with TVS protection on the inputs
- Insulation strength for input signals (opto-coupler characteristic) from the module's logic section is 3750Vrms
- Digital input voltage is up to 36VDC:
  - 0 to 2 V – opto-coupler closed, logical “1” on opto-coupler output
  - 4 to 36V – opto-coupler open, logical “0” on opto-coupler output
- Changing the state of digital input signals can be used to generate an interrupt (**INT** signal in the **HOST** connector).
- 2 optically isolated digital outputs
- Insulation strength (opto-coupler characteristic) for output signals from the logical section of the module is 3750Vrms
- Maximum current I_C for digital output channels is 8A (16A peak), with a voltage up to 100VDC, and dissipated power of 20W. The module can be used in extreme conditions only with appropriate cooling, including radiators
- If SMAJ48A TVS diodes are installed to protect against transient processes, the maximum voltage is up to 48VDC
- When working with digital output channels: logical “1” (2.5 to 5V) means the transistor is closed; logical “0” (0 to 0.5V) means the transistor is open
- Low-side and high-side load connection schemes can be used
- 2 **SPST (1 Form A)** type relays
- Switching Voltage: 250VAC/30VDC
- Max. switching current: 5A
- Coil resistance [±10%] (at 20°C 68°F) - 227 Ω
- Insulation resistance: 1000MΩ (at 500VDC)
- Breakdown voltage:
  - Between open contacts - 1000 Vrms for 1min
  - Between contact and coil - 3000 Vrms for 1min
- Surge breakdown voltage (Initial) (Between contacts and coil) - 6000 V

- When working with the module: logical “1” (2.4-5V) – this turns off the relay (NC)
  logical “0” (0-0.9V) – relay activation (NO)
- The module features LED indicators for the relay status: LED lights up when the relay is on
- The module features LED indicators for the status of digital input and output channels (the LED lights up when the opto-coupler on the input/output is open)
- Reading the input signal states and controlling the output signals are done using **AP0-AP7** (signals **IN1-IN4**, **RL1-RL2** and **DO1-DO2**) through the **HOST-P** connector and/or via the **I²C** bus through the I/O expander
- The I/O expander used in the module is the [**TCA9534**]() or a compatible chip
- The expander's **I²C** address (**A2-A0**) is set using DIP microswitch:

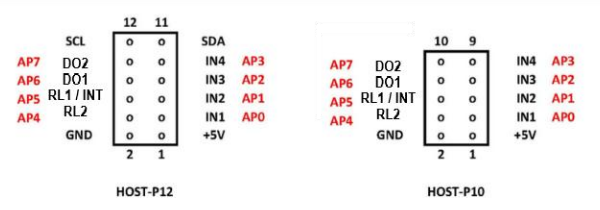
TCA9534	0	1	0	0	A2	A1	A0	x
TCA9534A	0	1	1	1	A2	A1	A0	x

- Connection to the module via the **I²C** bus is done through [**Qwiic©**]() connectors or through pins 11 (**SDA**) and 12 (**SCL**) of the **HOST-P12** connector
- Transient suppression and electrostatic discharge protection of signals on **Qwiic©** connectors is done using a TVS diode assembly (ESD protection)
- Module size is 47x56 mm. The module has mounting holes allowing it to be installed in the base module or in a Raspberry Pi

HOST-P CONNECTOR:

Depending on the version of the **IoTextra Octal** module can have either a 12-pin (**HOST-P12**) or a 10-pin (**HOST-P10**) **HOST-P** connector.

The pinouts for these connectors are shown in the image below:



The **HOST-P** connector is used differently depending on how the microcontroller interacts with the **IoTextra Octal** module:

1) ***Standalone Use:*** In this mode, the input signal states and control of the output signals are available via the **HOST-P10** (**HOST-P12**) connector or via the **I²C** bus using the I/O expander.
2) ***Smart Use:*** In this mode, an **IoTsmart** microcontroller module is vertically inserted into the **HOST-P12** connector. Reading the input signal states, control of the output signals and the 5V module power supply are done through the **HOST-P12** connector. Power is supplied from the **IoTsmart** module. Below is a photo of the **IoTextra Octal** module with the **IoTsmart** module:

IoTextra Octal with **IoTsmart ESP32-S3**

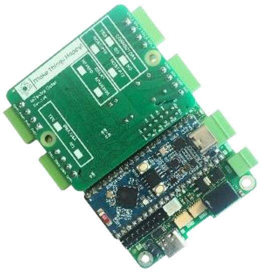
IoTextra Octal with **IoTsmart RP2350**

IoTextra Octal with **IoTsmart RP2040**

IoTextra Octal with **IoTsmart XIAO**

3) ***Mezzanine Use:*** In this mode, the **IoTextra Octal** module is installed in the base module, the state of the input signals and the output signal control are available through the **HOST-P12** connector. Photo of the **IoTextra Octal** module installed in **IoTbase** modules below.

IoTbase PICO with **PICO2** and **IoTextra Octal** modules

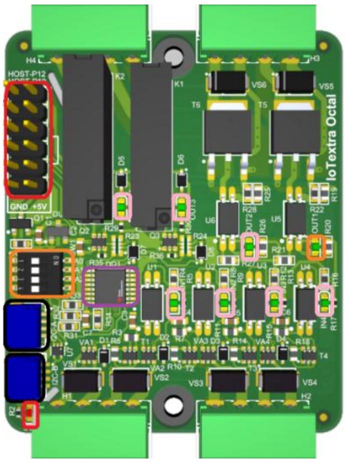


IoTbase PICO with **Waveshare ESP32-S3-Pico** and **Textra Octal** modules

IoTbase NANO with **Waveshare ESP32-S3-NANO** and **IoTextra Octal** modules

COMPONENT LAYOUT:

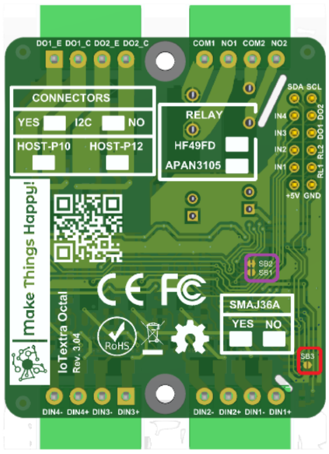
There are three versions of the **IoTextra Octal** module, and accordingly, each with a different **top-side** component layout:



It is possible to obtain input signal status and control output signals via the **HOST-P** connector (highlighted in blue).
The **Qwiic©** connectors are highlighted in black.
The I/O expander is highlighted in purple.
The DIP microswitch for setting the I/O expander's address on the **I²C** bus is highlighted in orange. The "ON" position of the microswitch corresponds to a "0" value in the respective address bit (A0, A1, and A2) for the I/O expander. The "OFF" position corresponds to a "1" value.
There is a 3V3 power indicator. The power indicator are highlighted in red in the **top-side** image.
The I/O signal indicators are shown in pink in the figure.

JUMPERS:

The image below shows the component and marking layout on the **bottom-side** of the **IoTextra Octal** module:



The **bottom-side** of the **IoTextra Octal** module has jumpers **SB1** and **SB2** for setting AP4. If **SB1** is closed, **AP4** is an **INT** from the I/O Expander. If **SB2** is closed, **AP4** controls relay **RL1**. By default, **AP4** is used to control **RL1**, and the interrupt is not used.
The **bottom-side** of the **IoTextra Octal** module has a jumper **SB3**, that allows powering devices connected through the **Qwiic©** connector. It is highlighted in red. By default, the jumper is absent.

CONFIGURATION TABLES:

The **bottom-side** of the module provides confiration information:

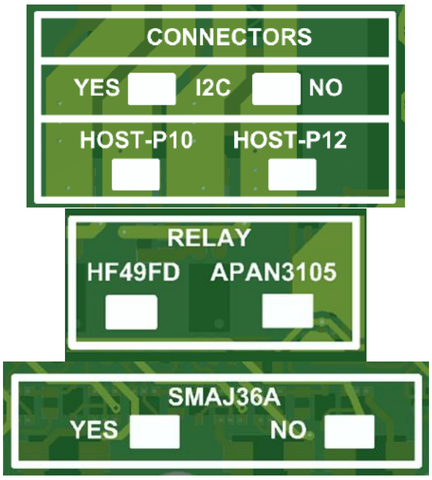
What connectors are installed in the module:

- **Qwiic©** connectors for **I²C**
- **HOST-P10**
- **HOST-P12**

What relays are used in the module:

- **Hongfa** **[HF49FD]**
- **Panasonic** **[APAN3105]**

TVS diodes **SMAJ36A** installed or not



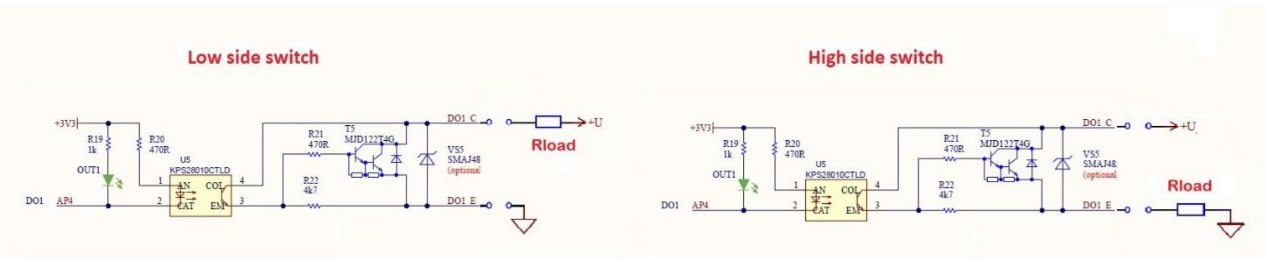
EXTERNAL SIGNAL CONNECTION:

Removable terminal blocks with a 3.5 mm pitch are recommended for connecting external input signals to the **H1-H2** connectors on the module. The pin layout is marked on the **bottom-side** of the module.

To connect external output signals, the **H3** connector on the module are used, into which it is recommended to insert removable pads with a pitch of 3.5 mm for terminal blocks. The location of the contacts is marked on the **bottom-side** of the module.

The relays are connected to connector **H4**.

The digital output channels are implemented using NPN Darlington Power Transistors, and the load can be connected using both the **Low Side Switch** and the **High Side Switch** schemes. For more details, see the following figure (for the DO1 channel):



ACCESSORIES:

The following accessories may be required for using the module:

- A set of four removable terminal blocks with a 3.5mm pitch for terminal blocks **H1-H4**
- A set of two standoffs and four screws for mounting the module into the **IoTbase** series base module.
- Cable for the **HOST-P10** connector
- Cable for the **Qwiic©** connector